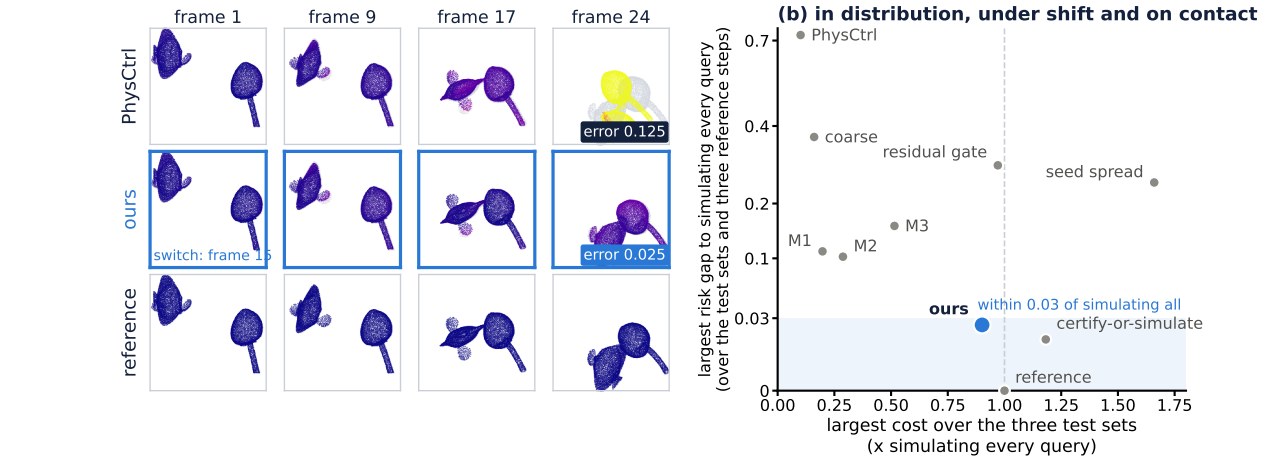

(a) two objects collide
frame 1
frame 9
frame 17
frame 24
PhysCtrl
ours
reference
error 0.125
switch: frame 15
error 0.025
(b) in distribution, under shift and on contact
largest risk gap to simulating every query
(over the test sets and three reference steps)
largest cost over the three test sets
(x simulating every query)
PhysCtrl
coarse
residual gate
seed spread
M1
M2
M3
ours
within 0.03 of simulating all
certify-or-simulate
reference
0.7
0.4
0.2
0.1
0.03
0
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75